

Cover Letter

Nishant Kheterpal

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Dear Search Committee,

I am writing to apply for the [POSITION HERE] position at [DEPARTMENT AND SCHOOL]. I am a final-year PhD candidate at the University of Michigan in the Robotics department, advised by Jean-Baptiste Jeannin. I have eleven semesters of university teaching experience across three institutions, two as instructor of record and nine as teaching assistant (TA). I began helping teach at the college level nine years ago, as a sophomore undergraduate at Berkeley, and have been a TA for four unique courses, spanning introductory programming, introductory and upper-level data science, and graduate-level robotics. Over Summer 2024, I was a lecturer at the Halicioğlu Data Science Institute at UC San Diego, where I taught two introductory data science classes, [DSC 10](#) and [DSC 40A](#), as instructor of record; I was responsible for all aspects of the course from staff hiring to lecture delivery to writing exams. Outside of my PhD research and coursework, I've taken advantage of opportunities to prepare myself for a teaching career. I have participated in programs like *Preparing Future Faculty* at Michigan and *WiscProf* at Wisconsin, and earned a Graduate Teacher Certificate from Michigan's Center for Research on Learning and Teaching. My teaching has been recognized at both Berkeley, where I won an Outstanding GSI award as an undergraduate, and at Michigan, where I was awarded an honorable mention (top 6 of 90) for the College of Engineering's Towner Prize for Outstanding GSIs; my course evaluations are consistently positive with above-average scores.

As a primary instructor at UCSD, I had the opportunity to teach two smaller-enrollment summer courses. I was able to apply what I've learned throughout the last nine years teaching to modify learning modalities when suited, mentor my own teaching assistants and support their growth, and offer summer instruction with the same rigor and quality as regular-year courses. The first course I taught, [DSC 10](#), covered Python programming and introductory statistics using simulation, and was originally based on [Data 8](#), a Berkeley course for which I spent five semesters as an undergraduate TA. I was able to leverage my past experience to, for example, prepare my own TAs to discuss nuanced statistics concepts like hypothesis testing. The second course, [DSC 40A](#), covered the mathematical foundations of learning from data: modeling, loss function optimization, linear algebra, and probability. I had not taught a class like DSC

40A before Summer 2024, but the topics covered lie at the core of fields like machine learning, control theory, and robotics.

I've mentored students in both formal and informal roles, in both teaching and research contexts. A past undergraduate mentee, Elanor, has since continued on to complete a master's degree in our research group and become a PhD student at Carnegie Mellon; we worked closely on the first two publications of my PhD. I've served as a more informal mentor and advocate for TAs, such as Ashley, a UCSD student who led discussions for my Summer 2024 DSC 10 class. Ashley is now the sole graduate student TA, leveraging her summer discussion experience in a larger academic-year role for which I recommended her. As a TA for [EECS 280](#), an introductory programming course at Michigan, I currently lead the content development and course-wide rollout of quizzes, a new course component, and have been involved in outreach as a moderator for the course's "Paths in Computer Science" panel. As the sole graduate TA for [Practical Data Science](#) at Michigan, I helped mentor our undergraduate TAs, all of whom were new to teaching discussions, via co-teaching, pedagogical discussions during office hours, and providing teaching feedback. I was involved in TA hiring for the Winter 2025 offering of *Practical Data Science*; I reviewed applications, conducted interviews including simulated office hours, and deliberated on candidates.

I'm drawn to [SCHOOL] for [REASONS]. I would be interested in teaching introductory courses like Data Structures ({{ number here }}), Algorithms ({{ number here }}), and Discrete Math ({{ number here }}); upper-division courses such as Machine Learning ({{ number here }}), Programming Languages ({{ number here }}), and Theory of Computation ({{ number here }}); or interdisciplinary courses in data science, robotics, and control theory. When relevant, I look forward to creating new courses, such as a {{ "foundational math course for machine learning that covers linear algebra, probability, and optimization" ONLY IF THE SCHOOL DOES NOT ALREADY HAVE ONE }}.

Thank you for your consideration. I look forward to hearing from you soon.

Best,
Nishant Kheterpal